

Approaches to Infectious Disease Modelling and Mitigation

Mathematical, statistical and AI perspectives

Prof. dr. Steven Abrams

**Workshop - Combining Infectious Disease Modelling and AI:
Methods and Communication**

September 14, 2026



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Simulation Models of Infectious Diseases



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INTRODUCTION



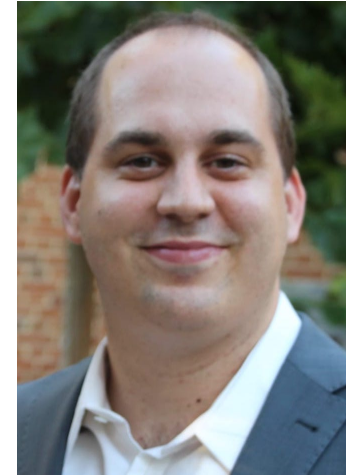
steven.abrams@uhasselt.be; steven.abrams@uantwerpen.be



steven-abrams-95752b43



<https://stevenabramslucp2479.github.io/>



- Associate Professor (Global Health Institute, University of Antwerp)
- Assistant Professor (Data Science Institute, UHasselt)
- **Main research activities:** mathematical and statistical modelling of infectious disease spread; focus on individual heterogeneity and serological data analysis

INTRODUCTION

WHO says challenges

Olivia Le Poidevin

By Reuters

PUBLISHED SEPTEMBER 8, 2026 AT 10:00

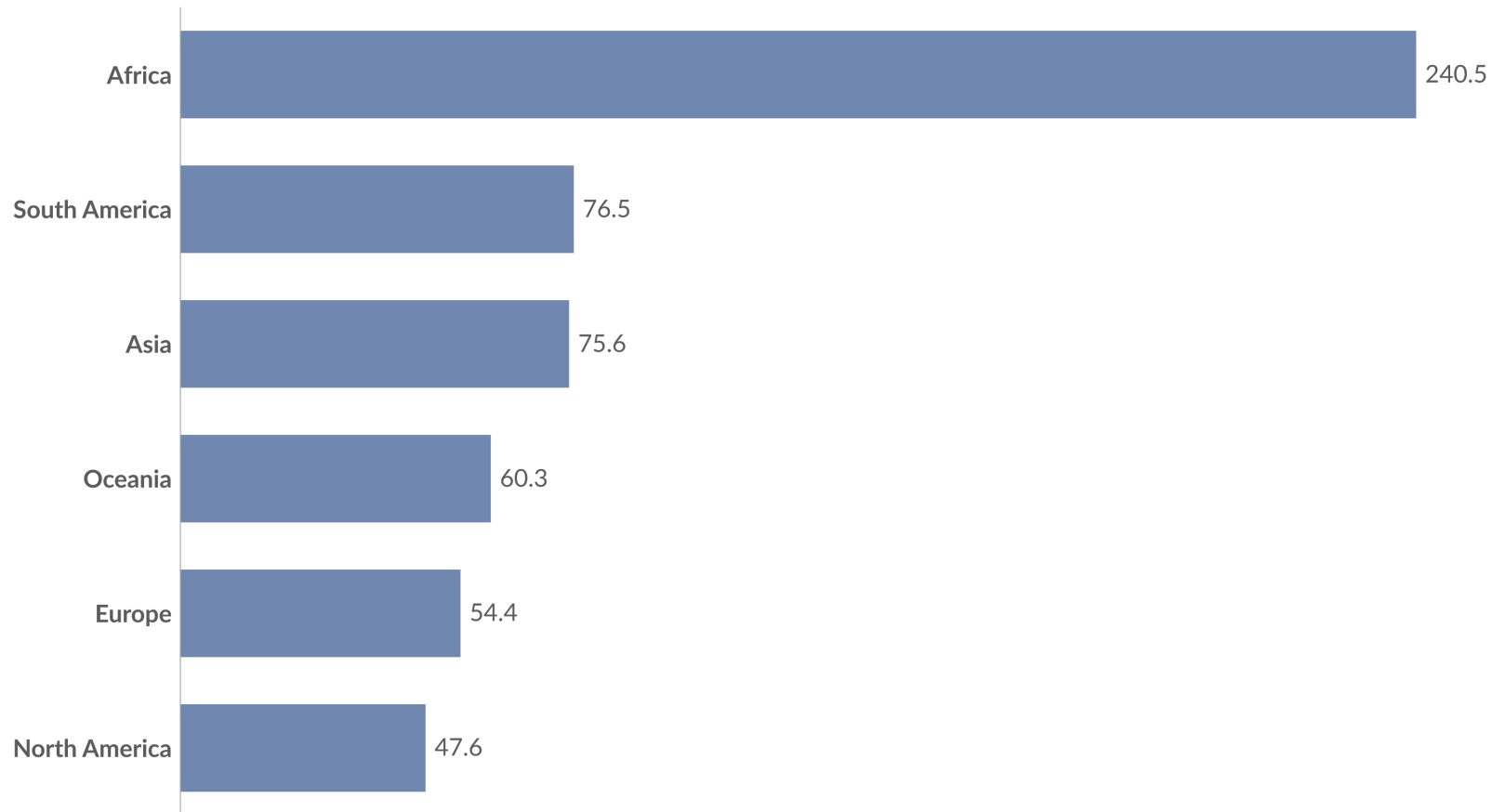


US CDC records more than 3,100 measles

Death rate from infectious diseases 2023

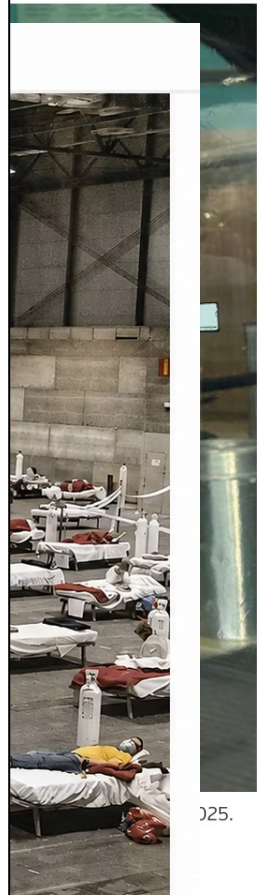
Our World
in Data

Estimated annual number of deaths from HIV/AIDS, diarrhea, malaria, respiratory infections, and all other infectious diseases per 100,000 people.



Data source: IHME, Global Burden of Disease (2025)

OurWorldinData.org/burden-of-disease | CC BY

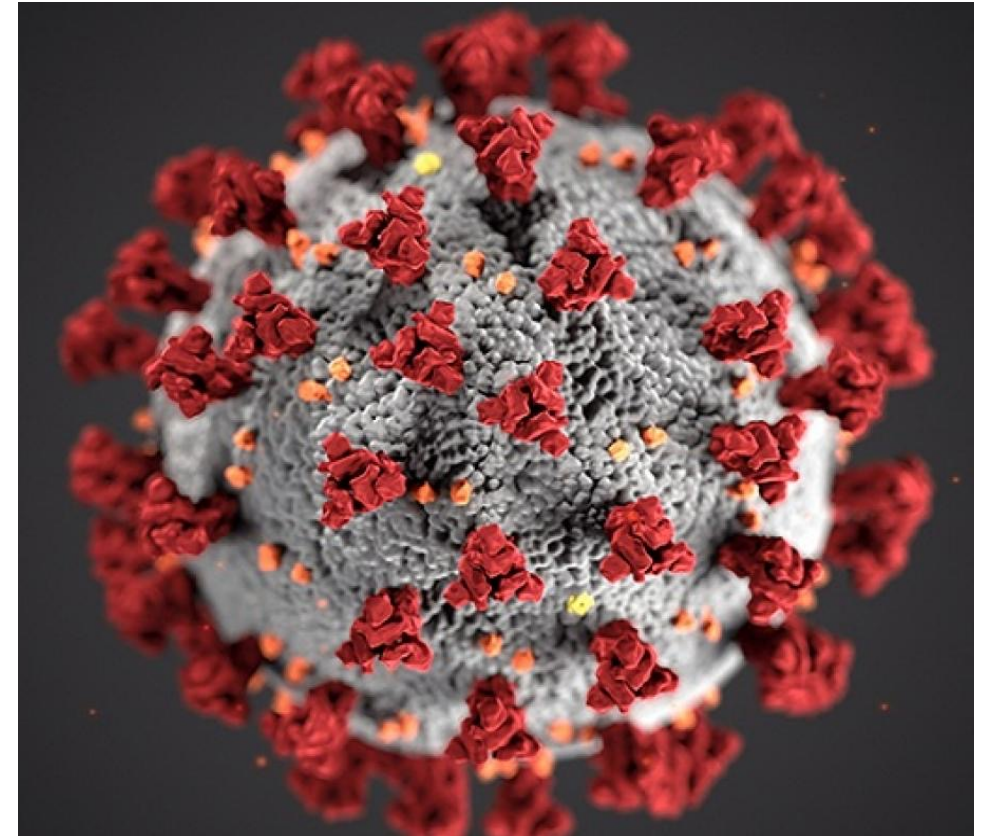


Infectious diseases

- There are countless of **infectious agents** that can infect human, animal and plant hosts
 - **Microscopically small infectious agents** with relatively short life spans, which replicate within their hosts are called microparasites such as **viruses, bacteria and fungi**
 - Larger infectious agents with relatively longer life spans called **macroparasites** such as **parasitic worms**
- **Transmission:**
 - **Directly** between hosts via respiratory air droplets or bodily fluids
 - **Indirectly** through an intermediary source, for instance via mosquitoes, ticks, rodents, environmental particles or contaminated blood products

Infectious diseases

- Infectious agents **evolve and transform** while new agents emerge regularly, implying their supply can be considered infinite
- Infectious agents causing disease in their hosts are often referred to as **pathogens**
- Infectious diseases are caused by pathogens, which are **transmissible between hosts**, either **directly or indirectly**



Mathematical, statistical and AI perspectives

Perspective	Main role	Typical representation
Mathematical modelling	Describe transmission mechanisms and express them mathematically	Compartments, contacts, networks, transitions between 'health' states
Statistical modelling	Learn from noisy, delayed and incomplete data	Likelihoods, latent states, uncertainty quantification
AI / machine learning	Learn flexible predictive patterns and learn optimal policies	Nonlinear functions, hybrid models, reinforcement learning

All approaches need assumptions, relevant data and evaluation (validation) of model outputs

Mathematical modelling frameworks

Infectious disease spread: how is the population represented?

Four frameworks at a glance

01 Compartmental models

Population is divided into broad subgroups (compartments). Individuals are tracked collectively as counts or proportions.



RANDOMNESS

Deterministic

Describes “average” population behaviour; the same inputs give the same trajectory.

Stochastic (probabilistic)

Chance determines how many individuals move between compartments; runs may differ.

TRANSMISSION FEEDBACK

Dynamic

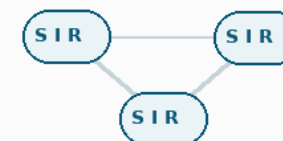
Infection risk responds to how much infection is in the population.

Static

Infection risk is specified externally, without transmission feedback.

02 Metapopulation models

Linked subpopulations, often places. Each can contain compartments; movement or mixing links them.



03 Network models

Individuals are nodes; contacts are edges. Transmission follows these connections. Captures who mixes with whom.



04 Microsimulation / agent-based models

Tracks individual attributes and histories. Microsimulation follows individual pathways; agent-based models can add interactions, behaviour and movement.



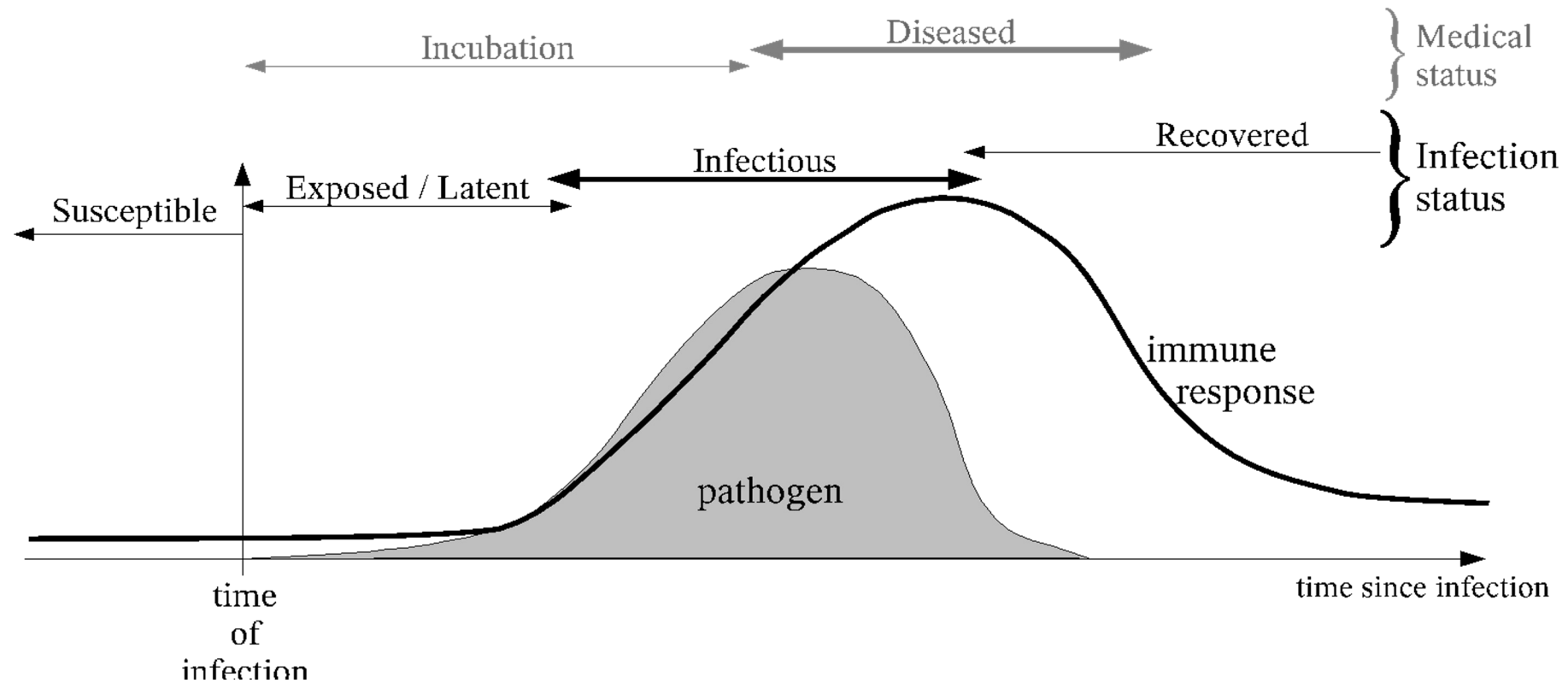
Frameworks can overlap: metapopulation models often use compartments; agent-based models can use networks.

Deterministic/stochastic and dynamic/static are separate modelling choices.

Illustrative schematics

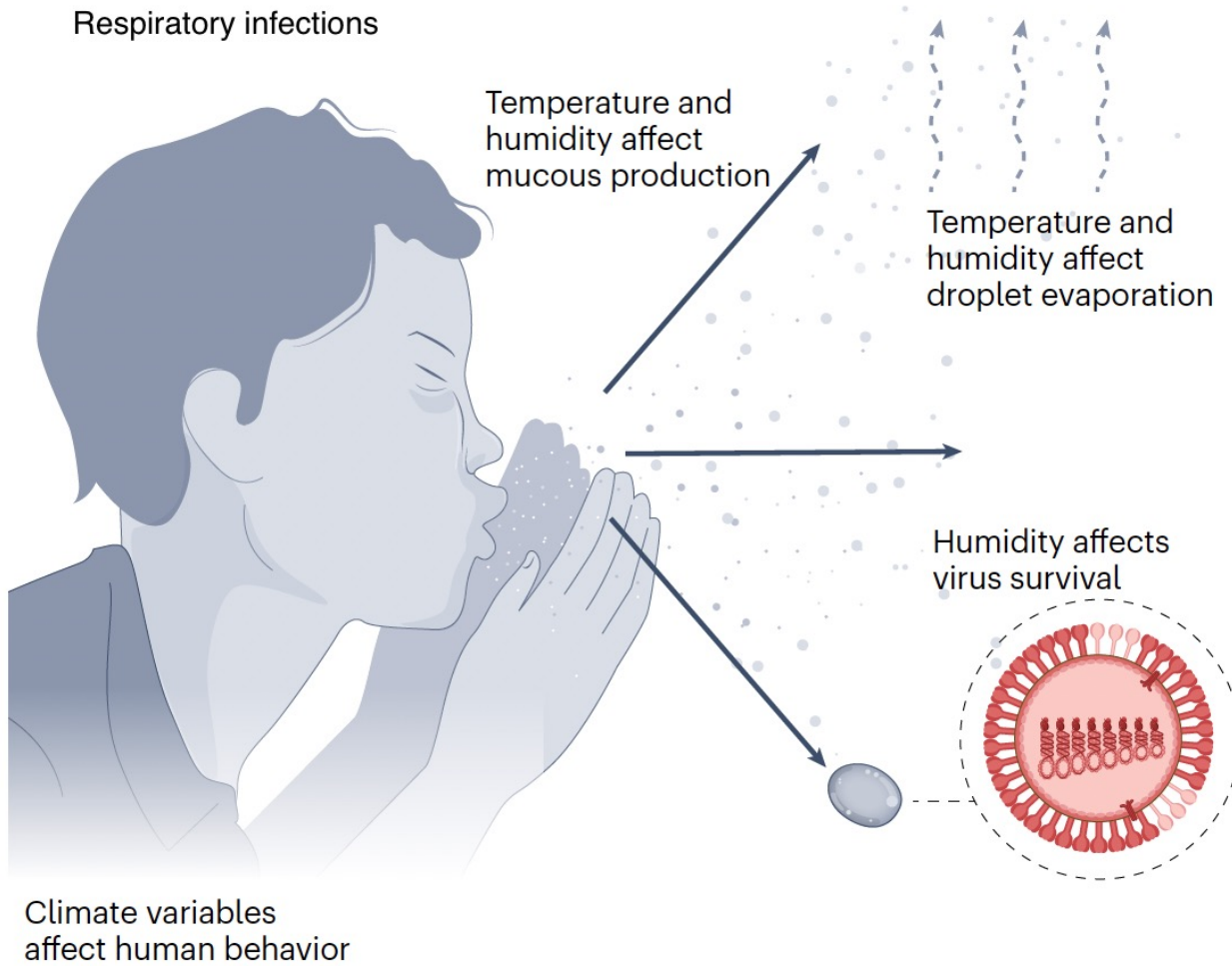
- **Compartmental models**
 - Population is subdivided into broad subgroups (compartments)
 - Individuals are tracked collectively
 - (Roughly) either deterministic or stochastic (probabilistic)
 - Deterministic models describe what happens 'on average' in a population
 - Stochastic models allow the number of individuals who move between compartments to vary through chance
 - Transmission dynamic or static models
- **Metapopulation models**
- **Network models**
- **Microsimulation or agent-based models**

Natural history determines model structure

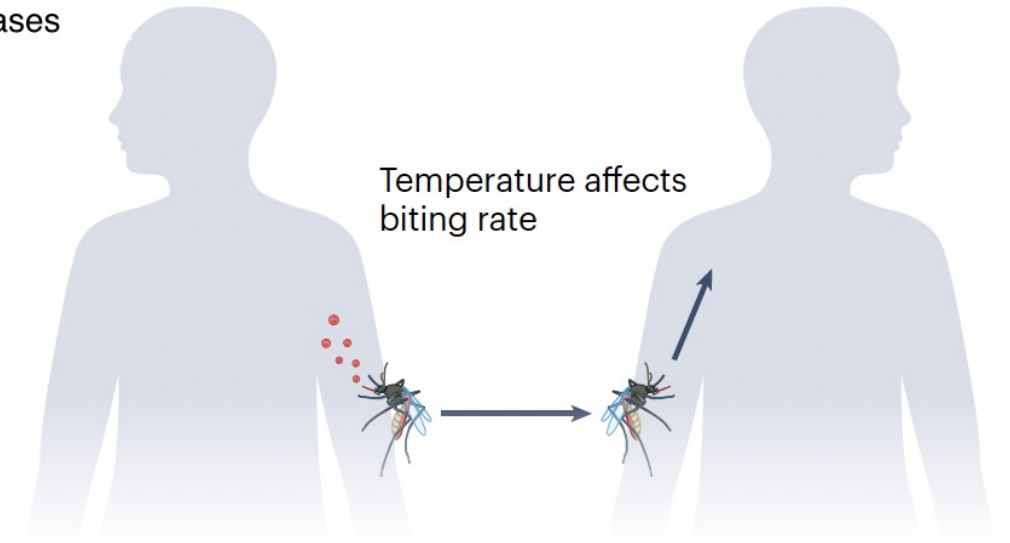


FOUNDATIONS – TRANSMISSION ROUTES

Respiratory infections



Vectorborne diseases



Source: Adapted from Baker et al. (2026).

- **Purposes:**
 - **Prediction:** requires the inclusion of known complexities and population-level heterogeneity
 - **Understanding:** investigating the factors that drive dynamics
- Building a model presents a **trade-off**:
 - **Accuracy:** reproduce what is observed and predict future dynamics
 - **Transparency:** ability to understand how model components influence the dynamics and interact
 - **Flexibility:** ease of adapting the model to new situations

MATHEMATICAL MODELS FOR INFECTIOUS DISEASE SPREAD

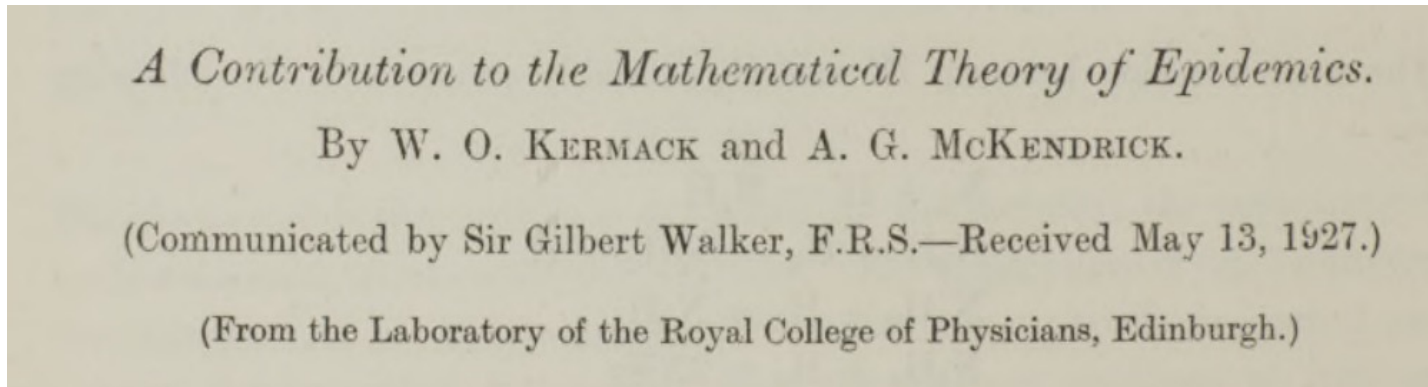
- Models are a **simplification of reality**
- Infectious disease transmission is driven by **chance events**, hence, this **hinders perfect prediction**
- **Good model:**
 - As simple as possible, but no simpler
 - Balance accuracy, transparency, flexibility
 - Parametrisable from available data

- The **principles**:
 - **1760: Daniel Bernoulli**
~ **Jean le Rond d'Alembert**
 - smallpox eradication
 - competing risks
 - McKendrick (1926): EM algorithm
 - Kermack & McKendrick (1927): the SIR model
 - Bartlett (1956): chance events
 - ...
- Many **seminal contributions** since then:
 - Bailey (1975)
 - Anderson and May (1992): 'The Bible'
 - ...



Daniel Bernoulli (1700 – 1782)

- **Kermack and McKendrick (1927):**

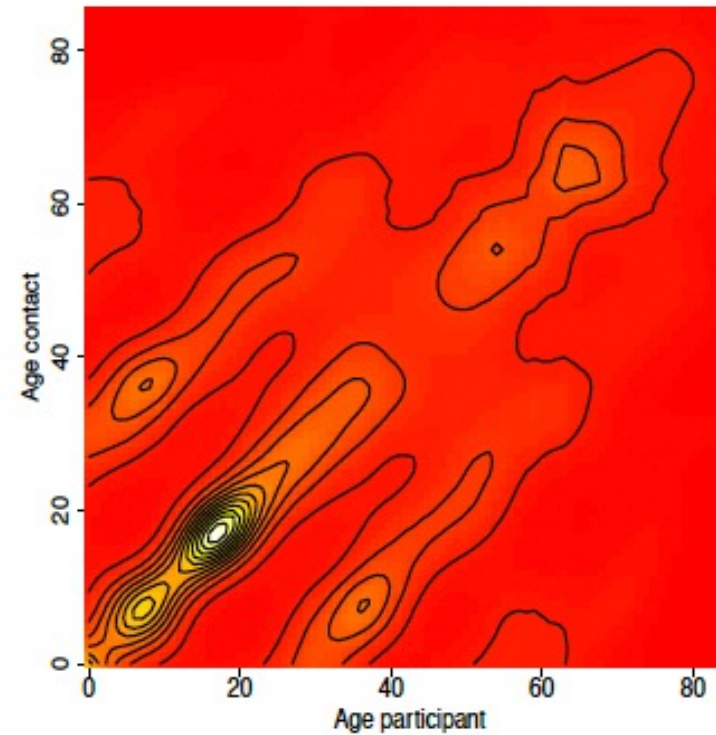
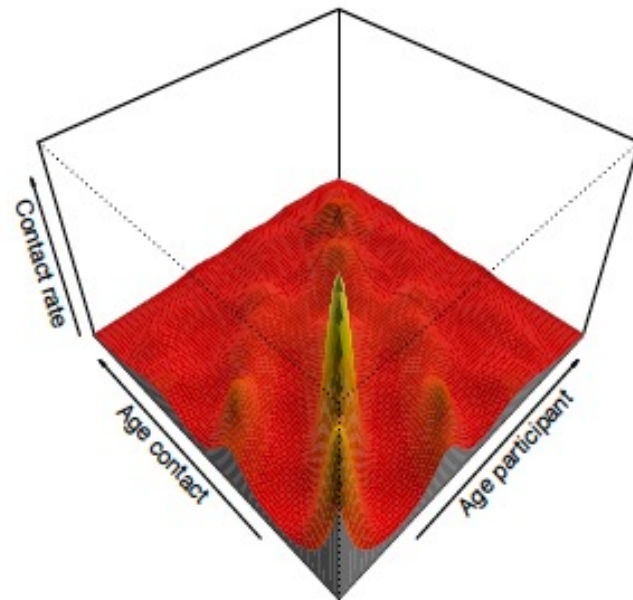


- **More recent contributions:**

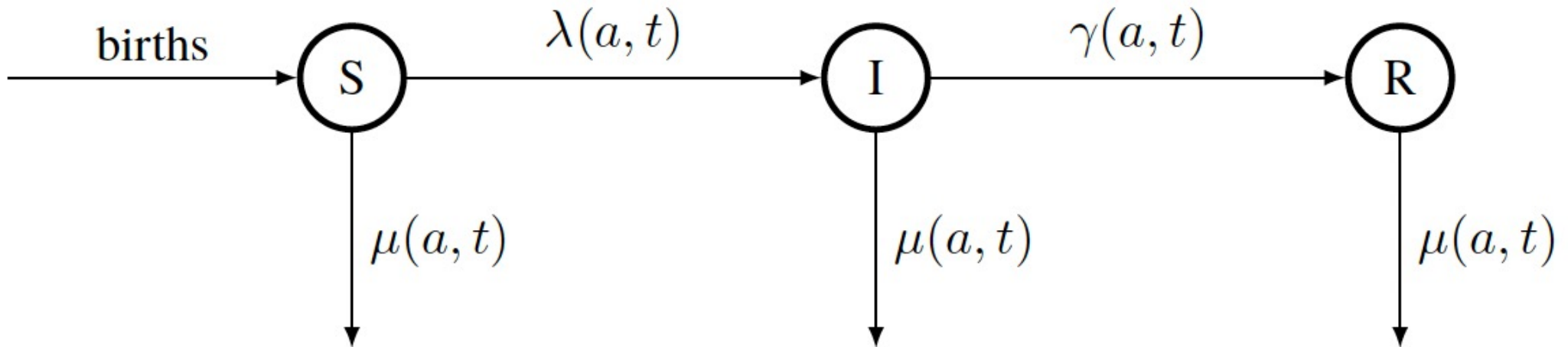
- Grenfell and Dobson (1995), Daley and Gani (1999), Hethcote (2000)
- Several **textbooks** are available:
 - Keeling and Rohani (2008)
 - Vynnycky and White (2010)
 - ...

PRINCIPLES OF MODELLING INFECTIOUS DISEASE SPREAD

- **How do people interact?** - contacts between individuals:



- Basics of compartmental models - **SIR model**:



- **Discrete time deterministic model:**
 - Transitions between states governed by so-called **difference equations**:

$$S_{t+1} = S_t - \lambda_t S_t$$

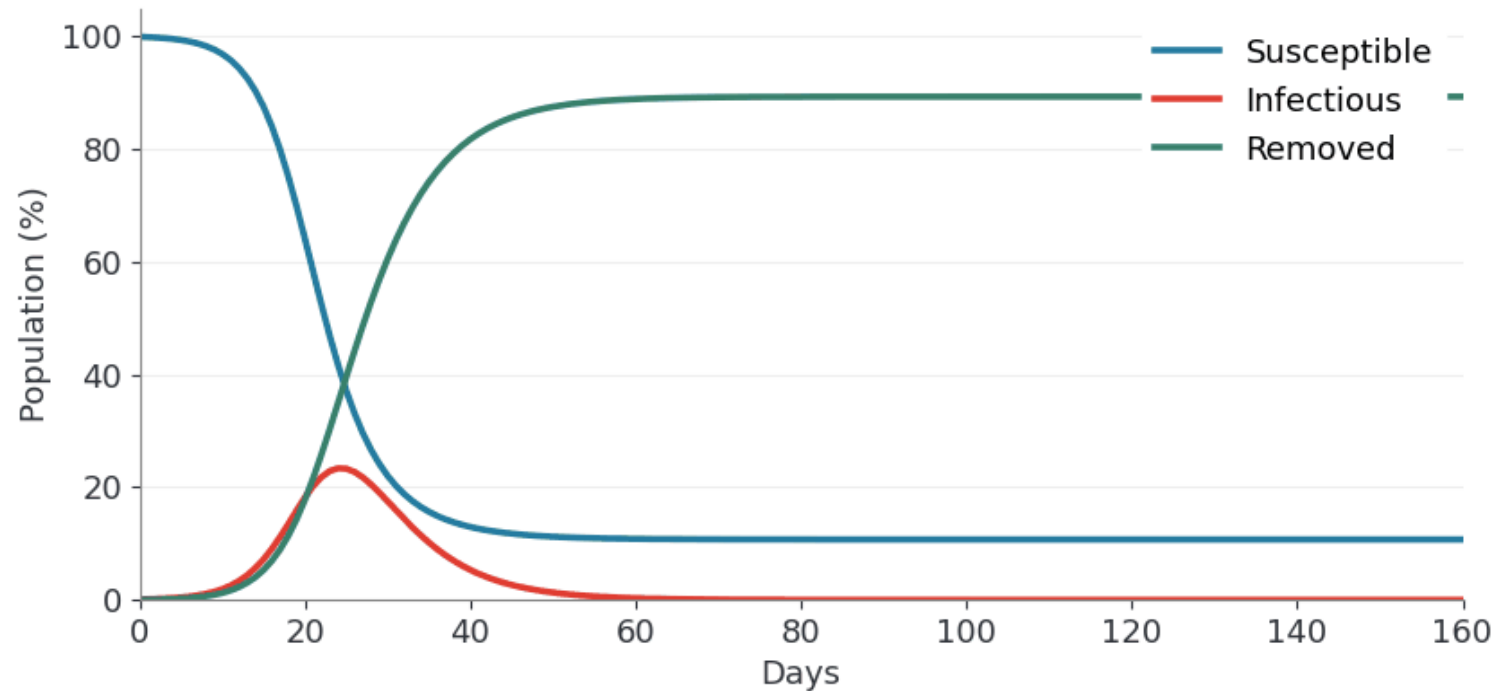
$$I_{t+1} = I_t + \lambda_t S_t - \nu I_t$$

$$R_{t+1} = R_t + \nu I_t$$

- **Continuous time deterministic model:**
 - Transitions described using a system of (ordinary) differential equations:

$$\begin{aligned}\frac{dS_t}{dt} &= -\lambda_t S_t \\ \frac{dI_t}{dt} &= \lambda_t S_t - \nu I_t \\ \frac{dR_t}{dt} &= \nu I_t\end{aligned}$$

SIR dynamics over an epidemic



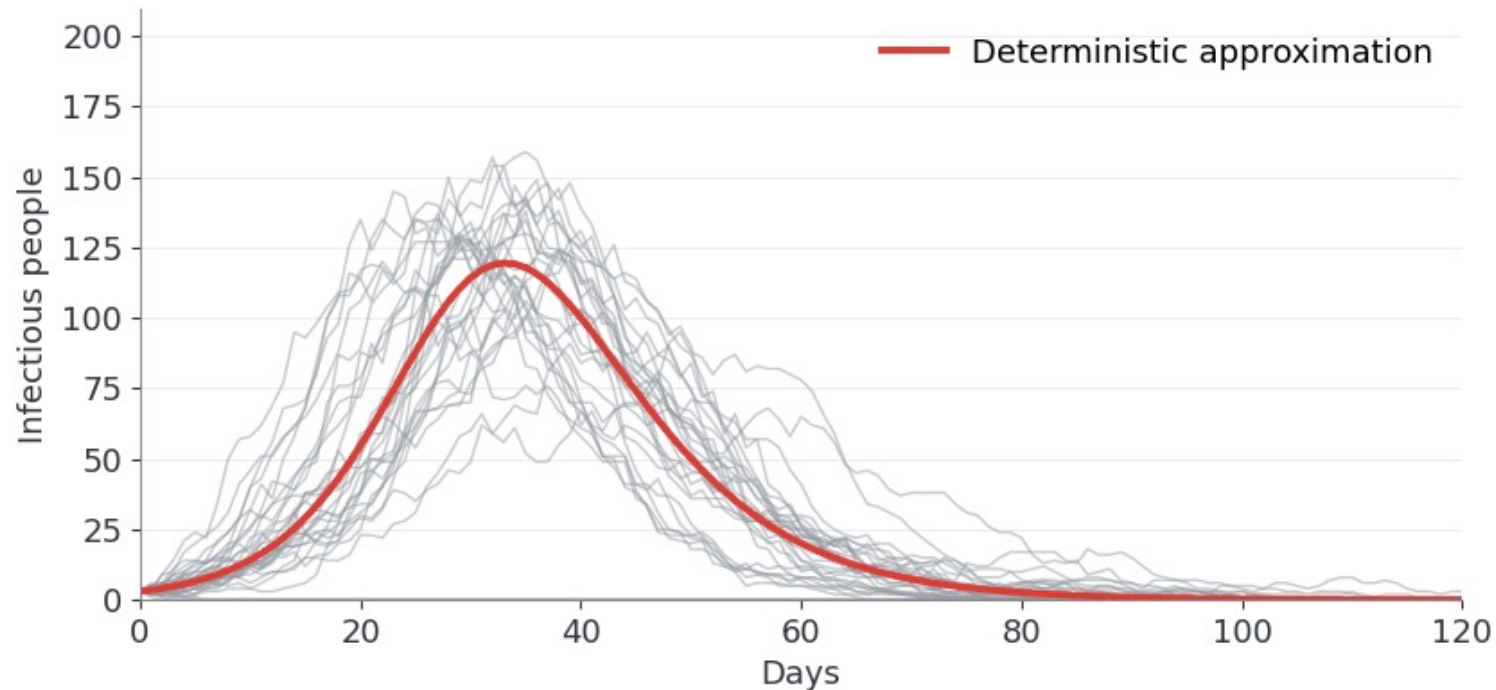
Illustrative simulation: the infectious population rises, peaks and declines

Stochasticity in disease transmission

- **Discrete time stochastic model:**
 - Transitions between states in a probabilistic manner, e.g. newly infected cases coming from e.g. a Poisson distribution with mean $\lambda_t S_t$

STOCHASTIC COMPARTMENTAL MODELS

Chance can produce fade-out or lead to a large outbreak

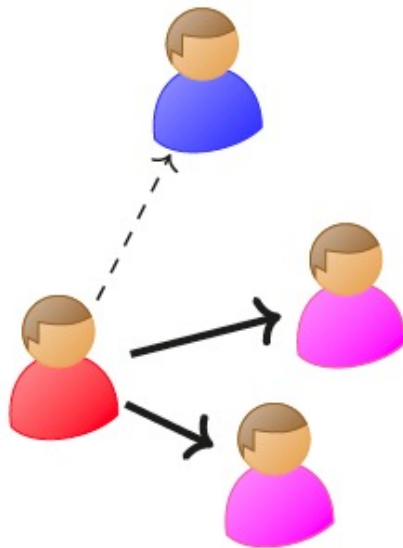


Illustrative stochastic SIR: identical parameters, different realised epidemics

PRINCIPLES OF MODELLING INFECTIOUS DISEASE SPREAD

- Basic reproduction number R_0 :

*The number of secondary cases caused by a single infected and infectious individual when introduced into a **fully susceptible population** during his/her infectious period*



Infectious Disease	R_0
measles	10-20
chickenpox	5-10
mumps	5-10
rubella	4-7
smallpox	3-5

Farrington (*Modelling Epidemics*, 2003)

- **Effective reproduction number R_t :**

*The number of secondary cases caused by a single infected and infectious individual when introduced into a **population with existing immunity** during his/her infectious period*

PRINCIPLES OF MODELLING INFECTIOUS DISEASE SPREAD

- Exponential growth (and decay)
 - $R_t > 1$ can lead to growth
 - $R_t \leq 1$ leads to extinction
- Without interventions epidemic spread cannot be mitigated unless there is a depletion of susceptibles
- **Heterogeneity** is key (superspreading events)
- Self-defeating prophecy or prophet's dilemma: “prevention paradox”

A statistical model links transmission to data

Latent epidemic process

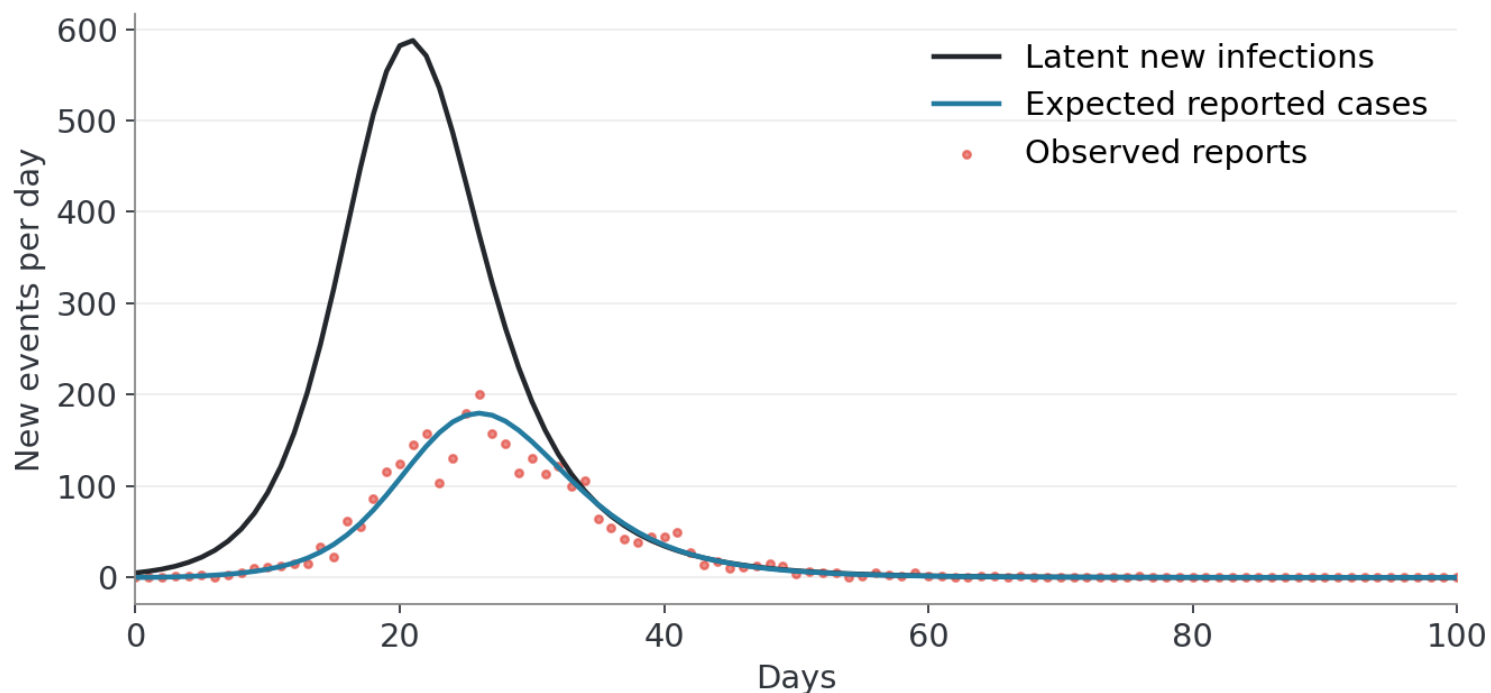
X_t depends on X_{t-1} , parameters and process variation

Observation process

Y_t depends on latent infections, detection, delay and measurement variation

Inference uses the observed Y to learn about the hidden X and unknown parameters.

Infections and reports follow different curves



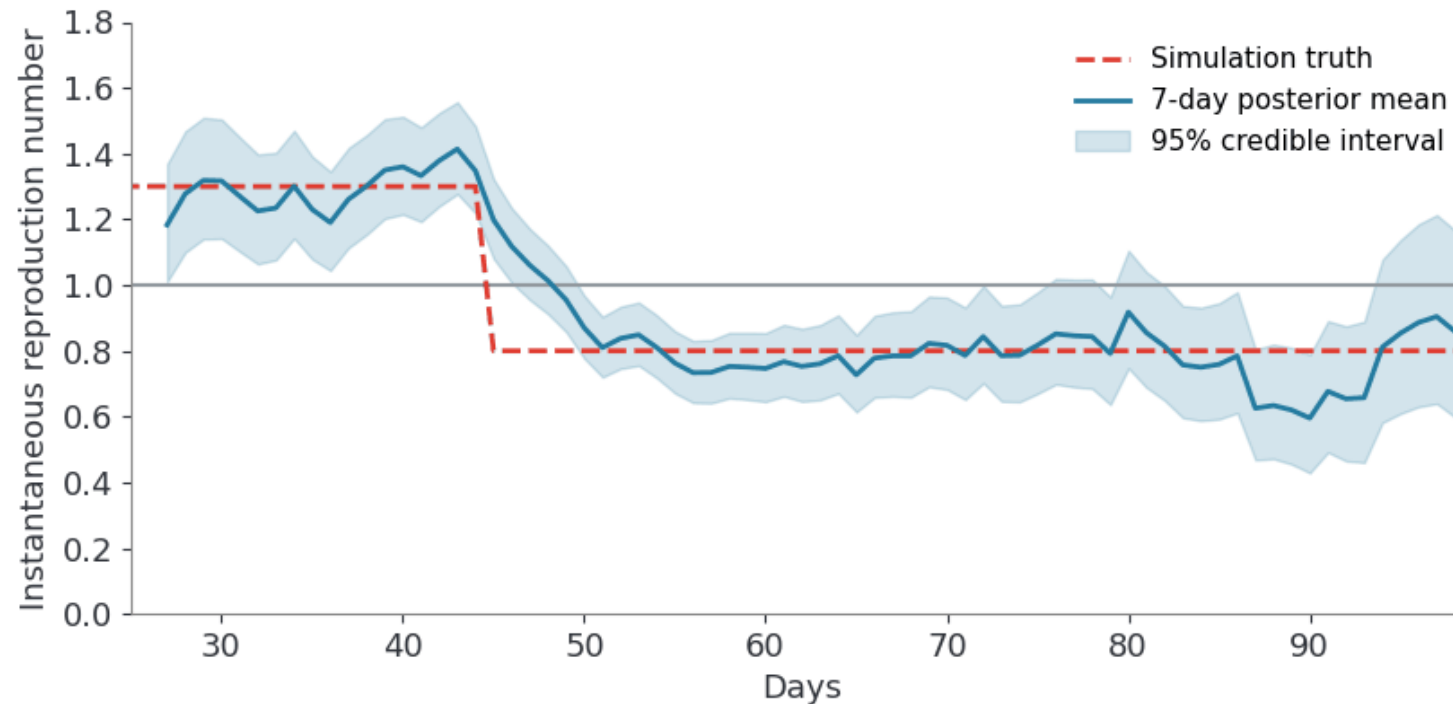
Simulated reports combine incomplete detection, a delay distribution and count noise

Likelihoods quantify uncertainty in counts

Model	Conditional variance	Typical use
Poisson: $Y_t \sim \text{Poisson}(\mu_t)$	μ_t	Counts with limited extra variation
Negative binomial: $Y_t \sim \text{NB}(\mu_t, k)$	$\mu_t + \mu_t^2/k$	Overdispersed surveillance counts
Binomial: $Y \sim \text{Binomial}(n, p)$	$np(1-p)$	Seropositive results among n tested

The likelihood should reflect how the data were generated and collected.

R_t estimates have uncertainty and lag



Synthetic infections with a known generation interval

Different data streams constrain different quantities

Data stream	What it can inform	Important limitation
Cases and admissions	Timing, trend and clinical burden	Ascertainment, delay and severity
Serology	Past exposure and immune markers	Sampling, vaccination and waning
Contacts and mobility	Mixing and behavioural change	Proxy for effective transmission
Wastewater and pathogen sequences	Community trends and lineage changes	Sampling, shedding and representativeness

Overview

Method	Statistical or mathematical approach	Main application
Growth models	Fit exponential, logistic and Richards curves to surveillance data	Detect trends and peaks Forecast about five days ahead
Transmission-tree analysis	Estimate serial and generation intervals from infector–infectee links and symptom onset	Characterise transmission timing and presymptomatic spread
Reproduction-number estimation	Relate epidemic growth to the mean and variance of the generation interval	Estimate transmission intensity Assess bias from serial-interval substitution

Source: Abrams et al. (2020). Nieuw Archief voor Wiskunde, 21(2), 84–89

www.nieuwarchief.nl/serie5/pdf/naw5-2020-21-2-084.pdf

CASE STUDY: SARS-COV-2

- SARS-CoV-2
 - **Person to person transmission** with a most likely zoonotic origin
 - **Respiratory infection**
 - **Pre-symptomatic** transmission
 - **Asymptomatic** transmission
 - Human mobility
- COVID-19
 - **Symptoms:** none, mild to severe
 - **Substantial mortality**
- **Transmission:**
 - Droplets & aerosol
 - Fomites less likely (exposure)
- $R_0 = 2-4$
- **Incubation period:**
 - 4-6 days on average
 - 95% percentile 14 days
- **Infectious period:**
 - Uncertainty remains
 - 3-5 days before SO, 5 days after

CASE STUDY: SARS-COV-2

Epidemics 35 (2021) 100449



Contents lists available at ScienceDirect

Epidemics

journal homepage: www.elsevier.com/locate/epidemics

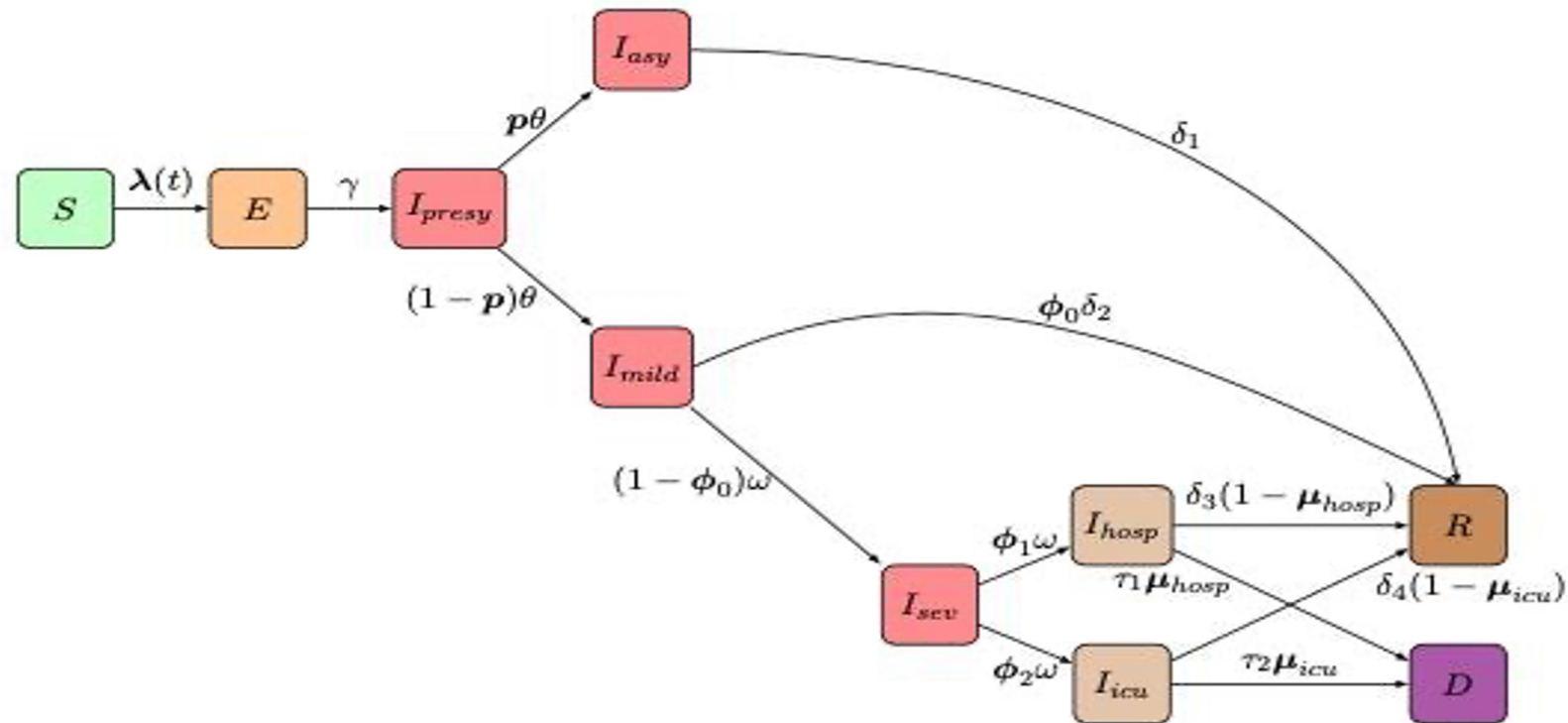


Modelling the early phase of the Belgian COVID-19 epidemic using a stochastic compartmental model and studying its implied future trajectories

Steven Abrams^{a,b,*}, James Wambua^a, Eva Santermans^a, Lander Willem^c, Elise Kuylen^c, Pietro Coletti^a, Pieter Libin^{a,d,e}, Christel Faes^a, Oana Petrof^a, Sereina A. Herzog^c, Philippe Beutels^c, Niel Hens^{a,c}

CASE STUDY: SARS-COV-2

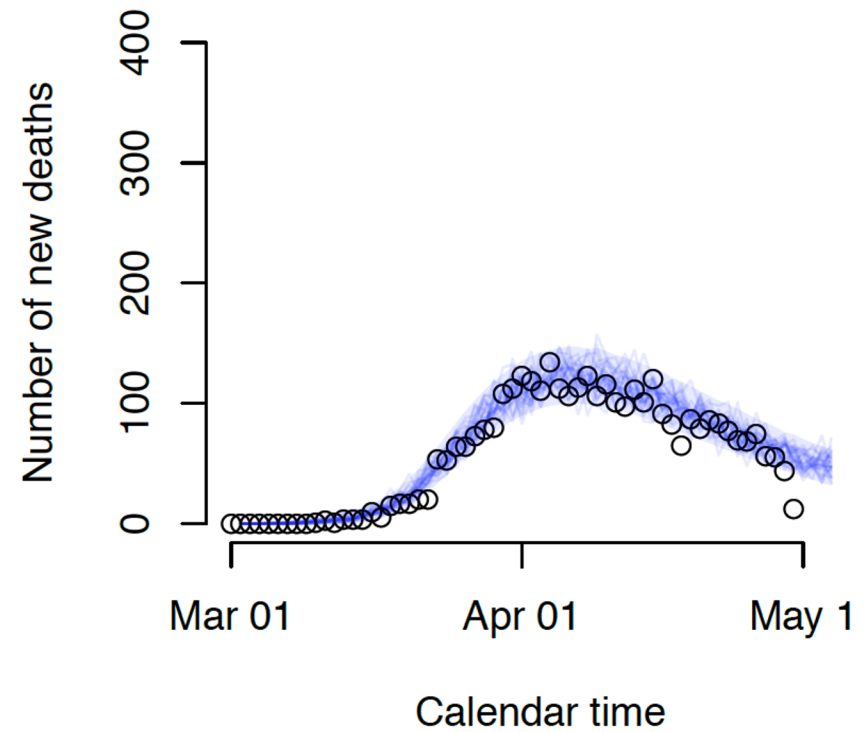
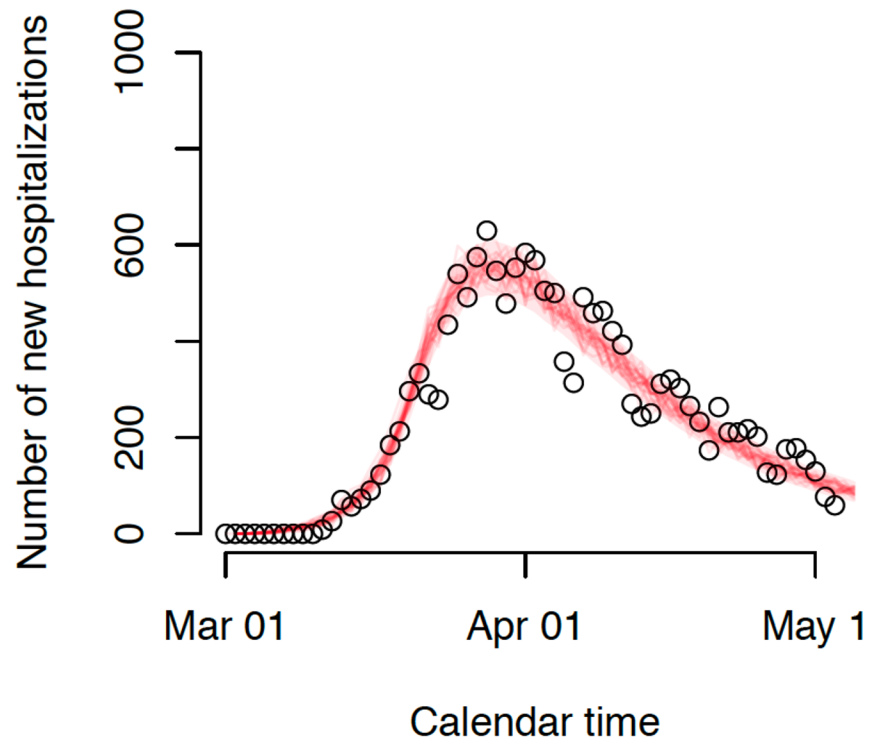
Schematic diagram of the stochastic model



Abrams et al. (2021)

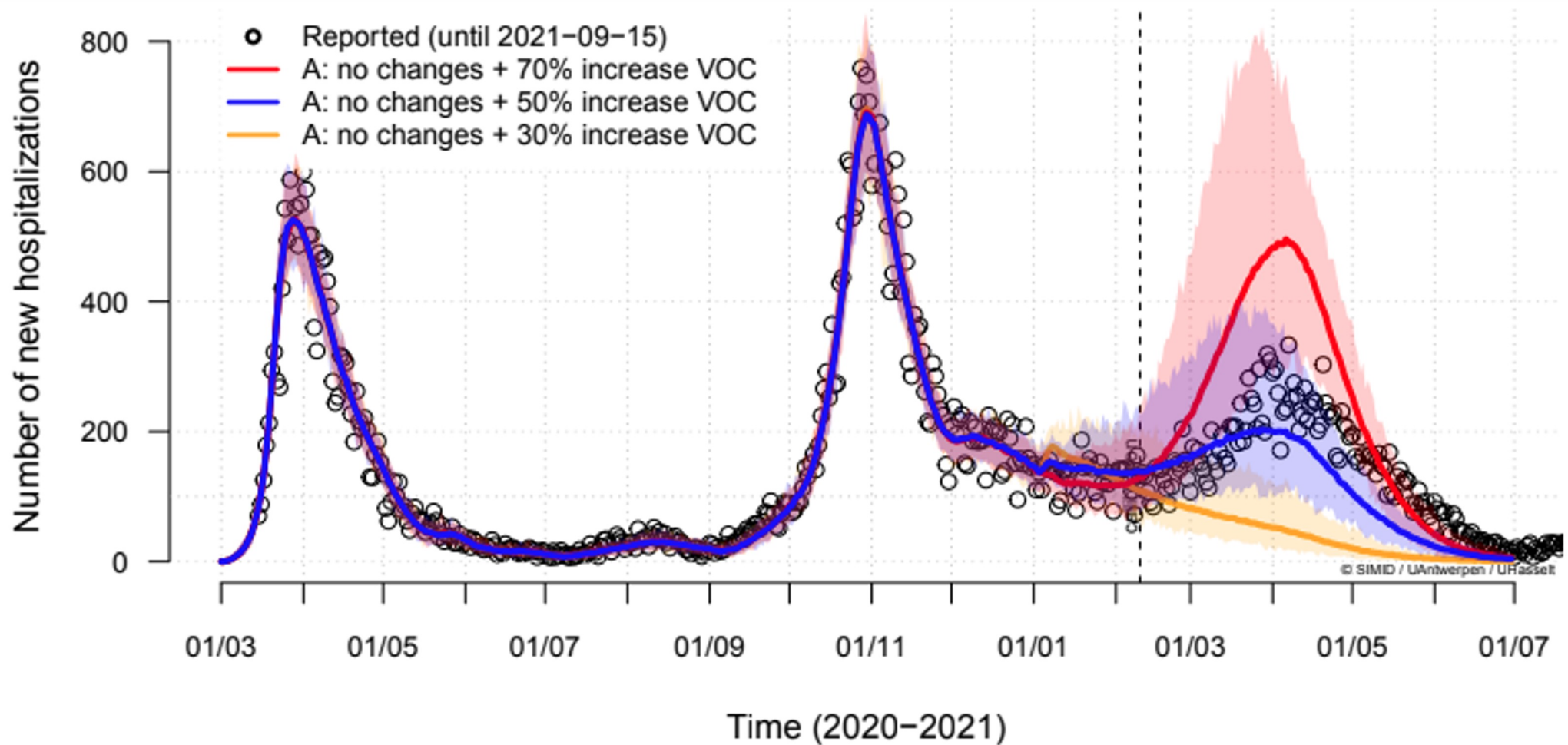
CASE STUDY: SARS-COV-2

Stochastic model: fitting to incidence data regarding hospitalizations and deaths

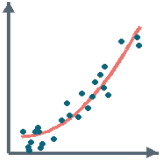
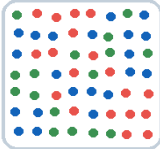
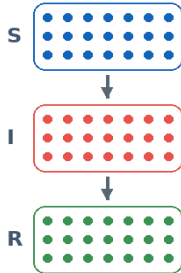


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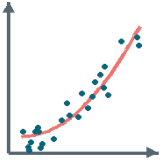
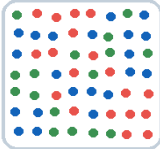
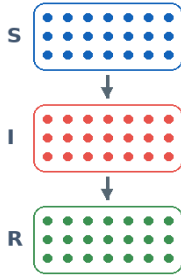
Press conference (26/02/2021):



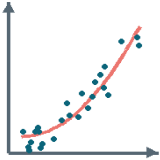
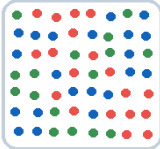
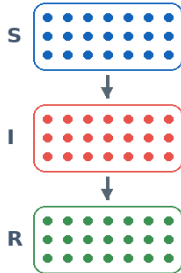
CASE STUDY: SARS-COV-2

Model / horizon	Input	Output	Main assumptions and limitations
Very short term Regression model	Indirect data from the past 2 weeks associated with hospital admissions. 	Estimated number of new hospital admissions over the next 2 weeks.	1. Indirect data potentially associated with hospital admissions (telecommunications data, absenteeism, test positivity ratio, etc.) are used as proxies. 2. Only useful as an early warning for hospital capacity.
Short term Individual-based model (IBM)	11 million interacting individuals, calibrated to hospital admissions over previous months. 	Estimated relative impact of NPIs on hospital admissions and R_e over the next 2 months.	1. Social contacts are used as a proxy for transmission events; different NPIs are therefore not distinguished. 2. Based on hospital admissions rather than infections, so the model lags behind by approximately 2 weeks. 3. Does not account for travel. 4. Does not account for weather conditions or seasonal variation. 5. Complex interactions among 11 million individuals; a supercomputer is needed for computation.
Long term Stochastic compartmental SEIR/PDE model (Reference model)	A population of 11 million divided into compartments (susceptible, exposed, infectious, recovered), calibrated to hospital admissions and deaths over previous months. 	Estimated relative impact of reopening sectors on hospital admissions and deaths over the next 6-7 months. Main model used, for example, for procurement by FAMHP.	1. Social contacts are used as a proxy (as in the IBM). 2. Pre-symptomatic, asymptomatic and symptomatic cases (mild and severe) have different levels of infectiousness; the proportion of asymptomatic cases is constant. 3. Immunity after infection. 4. No imported infections enter the population, despite, for example, 1.3 million PLFs being completed in August 2020. 5. Very limited information on hospital discharge; hospital capacity can therefore only be predicted indirectly. 6. Does not account for weather conditions or seasonal variation.

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Forecasts and scenarios answer different questions

	Forecast	Scenario
Question	What is likely next?	What if these assumptions hold?
Future behaviour	Modelled or averaged over	Specified as part of the scenario
Evaluation	Compare with future observations	Check assumptions and conditional implications
Example	Admissions next week	Admissions if contacts fall by 30%

CONCLUSIONS

Mathematical and statistical models alongside AI

Contribution	Potential role	Evaluation question
Mechanistic structure	Transmission and intervention scenarios	Are key mechanisms plausible?
Statistical inference	Link data to parameters and uncertainty	Does the likelihood reflect observation?
AI / machine learning	Flexible components and prediction	Does it improve prospective performance?
Ensembles	Combine complementary forecasts	Does it outperform simple baselines?

Key messages

- Mathematical transmission models explain how interactions generate epidemic dynamics
- Statistical models connect those mechanisms to imperfect observations
- *Mitigation comparisons depend on assumptions, timing and the outcome of interest*
- Useful forecasts need honest uncertainty and evaluation on future data

Key messages

- Model revision is part of scientific practice
- New evidence can change parameters: contacts, delays or severity
- New mechanisms can change structure: vaccination, immune escape or waning
- New surveillance practices can change the observation model

Selected reading: mathematical modelling

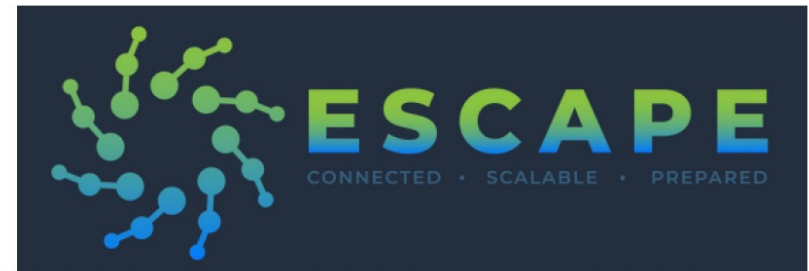
- Abrams et al. (2021). Modeling the early phase of the Belgian COVID-19 epidemic using a stochastic compartmental model and studying its implied future trajectories.
- Coletti et al. (2020). CoMix: comparing mixing patterns in the Belgian population during and after lockdown.
- Coletti et al. (2021). A data-driven metapopulation model for the Belgian COVID-19 epidemic: assessing the impact of lockdown and exit strategies.
- Franco (2021). COVID-19 Belgium: Extended SEIR-QD model with nursing homes and long-term scenarios-based forecasts.
- Kermack & McKendrick (1927). Mathematical theory of epidemics.
- Keeling & Rohani (2008). Modeling Infectious Diseases in Humans and Animals.
- Willem et al. (2021). The impact of contact tracing and household bubbles on deconfinement strategies for COVID-19.

Selected reading: inference and prediction

- Cori et al. (2013). Time-varying reproduction-number estimation.
- McGough et al. (2020). Nowcasting by Bayesian Smoothing.
- Held, Höhle & Hofmann (2005). Multivariate infectious-disease surveillance counts.
- Bracher et al. (2021). Evaluating epidemic forecasts in an interval format.
- Sherratt et al. (2023). European multi-model ensemble forecasts.
- Marshall et al. (2024). When are predictions useful?

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- FWO RESTORE project
- FWO ACCELERATE project (www.accelerate-epi.be)
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Philippe Beutels



Christel Faes



Geert Molenberghs



Andrea Torneri



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- CHERMID, Vaxinfectio, University of Antwerp
- GHI, University of Antwerp

Lander Willem



Pieter Libin



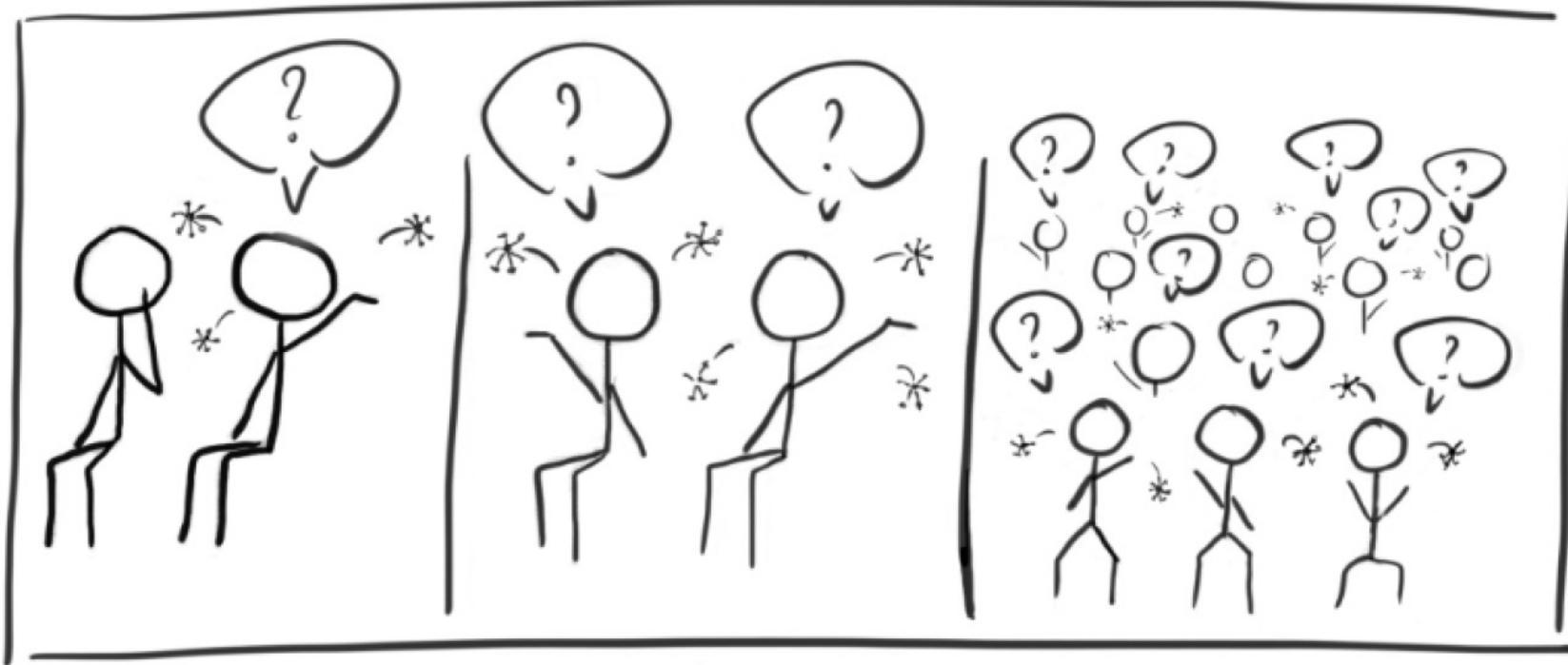
Patrick Mannion



Ilaria Misuri



Thank you! Questions?



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